Ex. No: 1

Perform footprinting on the target network using search engines, web services, and social networking sites

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1. Aim

To perform passive footprinting on a target network using search engines, web services, and social networking sites to gather publicly available organizational information.

2. Theoretical Background

Footprinting is the first phase of the ethical hacking process. The objective is to collect as much information as possible about the target system or network to identify paths for further analysis. Passive footprinting relies on open-source intelligence (OSINT) gathering without sending direct traffic or probes to the target's servers.

- **Search Engines:** Utilizing advanced operators (dorks) to filter results.
- **Web Services:** Using third-party sites like Netcraft or BuiltWith to analyze web technology stacks.
- **Social Networking:** Gathering organizational structure and employee details via LinkedIn, Twitter, etc.

3. Detailed Procedure

Step 1: Search Engine Dorking

Utilize Google Advanced Search operators to extract hidden directories, login portals, and sensitive documents.

- Search for general indexed pages: `site:rajalakshmi.org`
- Search for exposed PDF documents (brochures, internal guidelines): `site:rajalakshmi.org filetype:pdf`
- Search for administrative login pages: `site:rajalakshmi.org inurl:admin | inurl:login`

- Search for exposed directory listings: `site:rajalakshmi.org intitle:"index of"`

Step 2: Analyzing Web Services

Use third-party platforms to identify the technological footprint of the target.

- Navigate to a technology profiler (e.g., BuiltWith or Wappalyzer).
- Enter the target domain (`rajalakshmi.org`).
- Review the frameworks, web servers, and scripting languages in use.
- Use `crt.sh` to analyze SSL transparency logs and uncover subdomains.

Step 3: Social Networking OSINT

Explore publicly available social media profiles to map out human resources.

- Search the organization on LinkedIn to find the standard email formatting.
- Identify key personnel (IT Admins, Network Engineers).

4. Observation & Outputs

Category	Query / Method	Sample Extracted Output
Indexed Pages	`site:rajalakshmi.org`	Home, Admissions, Departments, Placements, Library sections indexed.
Exposed Documents	`site:rajalakshmi.org filetype:pdf`	Found: Admission_Brochure.pdf, Academic_Calendar.pdf
Admin Portals	`site:rajalakshmi.org inurl:login`	Located student/staff portal login pages.
Tech Stack	BuiltWith Profiler	Server: Apache Language: PHP, HTML5 Framework: Bootstrap
Subdomains	SSL Certificate Search (crt.sh)	www.rajalakshmi.org, mail.rajalakshmi.org
Social Media	Company Profile Search	Official LinkedIn page and public employee/alumni profiles identified.

5. Result

Passive footprinting was successfully performed on the target domain using search engines, web services, and social media. Information such as domain details, technologies, and public organizational information was successfully collected.